

STATESVILLE, NC, USA

WMO: 723055

Lat: 35.765N

Lon: 80.957W

Elev: 965

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	22.1	-1.7	5.2	27.9	3.3	6.7	29.6	18.7	43.2	16.7	43.2	2.9	300	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	92.0	73.9	90.3	73.5	88.2	72.8	78.0	85.1	76.6	84.5	75.4	83.5	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	141.9	80.3	74.6	134.2	79.6	73.0	127.1	78.7	42.2	85.8	40.7	84.4	39.5	83.6	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	12.8	11.2	11.0	95.0	4.9	3.2	7.5	97.2	4.6	99.1	1.9	100.9	-1.6	103.2		
			9.3	79.4	4.8	1.9	5.9	80.8	3.1	81.9	0.4	83.0	-3.1	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	39.0	42.6	50.4	59.4	67.1	74.6	77.1	76.2	70.6	59.5	49.2	42.9
	DBStd	15.33	9.66	9.06	9.26	8.04	6.48	4.48	3.85	4.38	5.88	7.94	8.15	8.49
	HDD50	1111	361	240	112	17	0	0	0	0	0	13	111	257
	HDD65	3537	806	627	457	202	58	2	0	0	18	205	475	687
	CDD50	4447	20	34	125	298	531	737	841	811	619	308	87	36
	CDD65	1397	0	1	5	32	124	289	376	346	187	35	1	1
	CDH74	11472	2	3	52	341	1009	2459	3162	2758	1364	301	19	2
	CDH80	4148	0	0	3	59	263	933	1296	1072	458	64	0	0
Wind	WSAvg	3.9	4.4	4.8	5.1	5.0	3.9	3.3	2.8	2.8	3.5	3.6	3.8	3.8
Precipitation	PrecAvg	45.4	3.6	3.6	4.4	3.4	3.9	4.4	3.9	3.9	3.8	3.4	3.2	3.7
	PrecMax	61.3	9.3	7.1	8.7	8.7	8.4	10.2	7.8	10.7	10.9	12.6	6.3	7.4
	PrecMin	27.3	0.8	0.6	1.3	0.4	0.9	0.8	0.8	0.6	0.0	0.0	0.4	0.4
	PrecStd	6.8	1.8	1.6	2.0	1.9	2.0	2.2	1.9	2.0	2.6	2.5	1.5	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.2	73.4	79.4	84.9	89.4	94.3	94.4	96.8	92.4	86.5	76.5	72.2
		MCWB	58.3	61.0	60.0	63.5	69.8	73.7	74.1	75.0	71.6	71.2	65.5	61.1
	2%	DB	63.6	68.0	74.6	81.4	85.7	91.1	91.3	91.1	89.1	80.9	72.0	66.1
		MCWB	55.4	55.8	59.5	63.8	69.2	72.8	74.7	74.7	71.0	67.2	59.1	61.9
	5%	DB	59.8	63.7	71.3	78.4	82.7	88.5	90.0	89.5	85.7	77.1	68.3	61.8
		MCWB	51.7	54.2	58.1	62.7	68.0	72.1	74.5	73.6	71.0	65.7	59.8	55.3
	10%	DB	54.9	59.0	66.2	74.4	80.2	85.6	87.7	86.1	81.7	73.1	64.1	57.0
		MCWB	48.4	49.6	55.2	60.7	66.8	71.6	73.6	73.6	69.8	64.1	56.8	51.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.6	64.3	65.0	71.2	73.5	77.2	83.7	81.6	76.0	73.9	68.4	66.2
		MCDB	66.7	69.3	72.9	78.8	82.1	85.6	89.4	86.1	84.2	79.8	72.7	67.6
	2%	WB	57.9	60.4	62.6	68.0	71.7	75.5	78.2	78.1	74.5	71.6	64.8	61.9
		MCDB	61.5	65.5	70.3	76.3	81.5	84.9	85.0	85.3	82.7	77.5	68.3	66.1
	5%	WB	53.9	56.5	60.2	65.6	70.3	74.5	76.8	76.8	73.2	69.1	61.9	57.5
		MCDB	57.5	62.1	67.6	73.2	79.1	83.9	84.8	84.2	80.7	74.6	65.9	60.6
	10%	WB	49.3	50.9	57.8	63.2	68.7	73.2	75.4	75.4	71.9	66.1	58.0	52.5
		MCDB	54.4	56.6	64.5	70.9	76.5	82.3	83.7	82.5	78.8	71.0	63.1	55.5
Mean Daily Temperature Range	5% DB	MDBR	19.5	20.0	21.1	22.2	20.8	20.0	18.8	18.2	18.5	21.7	22.2	18.5
		MCDBR	24.6	25.9	25.3	26.2	24.4	24.0	22.0	22.1	23.6	24.8	26.9	22.7
		MCWBR	17.1	17.1	13.8	12.8	10.8	8.5	8.1	7.7	8.7	12.6	16.9	16.0
	5% WB	MCDBR	19.7	22.2	21.7	21.8	20.3	20.3	19.3	18.5	18.7	19.9	20.7	18.5
		MCWBR	16.5	18.0	14.5	12.7	10.0	8.8	10.5	9.7	9.2	12.2	17.0	15.0
Clear-Sky Solar Irradiance	taub	0.308	0.319	0.353	0.400	0.451	0.481	0.530	0.519	0.419	0.372	0.323	0.308	
	taud	2.499	2.463	2.405	2.276	2.161	2.118	2.007	2.016	2.286	2.399	2.496	2.530	
	Ebn at Noon	284	293	291	280	266	258	244	244	265	270	275	277	
	Edh at Noon	26	30	35	42	48	50	56	54	39	32	26	24	
All-Sky Solar Radiation	RadAvg	811	1018	1356	1692	1877	2006	1870	1718	1453	1190	901	695	
	RadStd	52	120	93	133	152	144	127	83	132	148	75	64	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+266	+171

Nomenclature: See separate page